

Analytic theory of two-dimensional NMR in systems with coupled macro- and micropores

David Linton Johnson* and Lawrence M. Schwartz

Schlumberger Doll Research, One Hampshire Street, Cambridge, Massachusetts 02139, USA

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Two-dimensional (2D) nuclear magnetic resonance (NMR) experiments involve a sequence of longitudinal (T_1) and transverse (T_2) measurements. When such experiments are applied to porous media, they are believed to provide new and important information regarding diffusive coupling between distinct pore subpopulations. However, we show in this paper that, in many cases of interest, this is simply not true. One often encounters systems in which the one-dimensional (1D) T_1 and T_2 processes are each controlled by just two distinct decay modes. If these modes form a complete set, then one can derive analytic formulas that describe, exactly, the 2D NMR measurements. Therefore, for such systems, the 2D measurements bring *no additional information* over that which is already present in the 1D results. Our predictions agree quite well with numerical results based on the microporous grain-consolidation (μ – GC) model.

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I. INTRODUCTION

Two-dimensional (2D) nuclear magnetic resonance (NMR) experiments have recently been applied to a wide range of fluid filled porous media including reservoir rocks [1–4], cements [5], catalysts [6], and surfactants [7]. These measurements provide a correlation between the initial (encoding) and final (detection) part of the experiment. Of particular interest is the T_2 - T_1 - T_2 (also known as T_2 -storage- T_2) protocol in which the spins are first rotated to be perpendicular to the static magnetic field, $\mathbf{B}_0$, where they undergo transverse (T_2) relaxation for a variable amount of time, τ_1 . Next, they are rotated to be parallel to $\mathbf{B}_0$ where they undergo longitudinal (T_1) relaxation for a fixed amount of storage time, t_s . Finally, the spins are rotated back to the transverse plane and undergo transverse relaxation for a second variable amount of time, τ_2 [3,5,8]. Typically for each value of the storage time, t_s , the NMR data are processed to produce lifetime distributions $\mathcal{P}(t_s; T_2^a, T_2^b)$ where $T_2^{a(b)}$ refers to a decay time relevant to the first (second) T_2 part of the experiment.

A T_2 -storage- T_2 example, based on microporous grain-consolidation (μ – GC) simulations [8] described below, is shown in Fig. 1 together with the corresponding 1D T_2 distribution. In the 2D cross-plot off-diagonal peaks are clearly evident. These peaks have been interpreted as providing new information regarding diffusive coupling between distinct spin populations (e.g., micro- and macropores in the μ – GC model) [8]. Indeed, by monitoring the amplitude of these off-diagonal peaks as a function of the storage time, t_s , several authors have estimated exchange rates or coupling parameters [3–5]. While this approach is intuitively appealing, we will show that the behavior of the off-diagonal peaks can sometimes be predicted from the one-dimensional (1D) lifetime distributions. In such cases no *new* information is contained in the T_2 - T_2 cross-plot. Thus, we maintain, the very significant complexity of doing the 2D experiments as opposed to the more conventional 1D experiments (CPMG for T_2 plus inversion recovery for T_1 [1–7]) is often unwarranted.

We consider here the simple but important examples of 2D NMR, namely those for which the relevant 1D relaxation can be described as the sum of two exponential functions. (This includes, as a subset, the two-site exchange model which has often been used to interpret 2D NMR data [5,8].) The analytic theory we develop allows us to predict, with no additional information, the integrated intensity of each of the peaks appearing in two-dimensional cross-plots of the kind shown in Fig. 1. Therefore, for such systems, the 2D measurement brings no additional information over that which is already present in the 1D results. Equally important, our analysis points the way forward if we are to understand the value of 2D NMR in more complicated systems. We present numerical results on a system in which three modes are needed to describe the 1D lifetime distributions. Our analysis can be extended to such systems and should allow us to identify which features of the 2D distributions contain information that is not already present in the 1D spectra.

II. BASIC CONSIDERATIONS

A. Analysis of 1D and 2D NMR measurements

In both 1D and 2D NMR, the evolution of the magnetization (magnetic moment per unit volume), $M(\mathbf{r}, t)$, is described by the diffusion equation [9]

$$D_0 \nabla^2 M(\mathbf{r}, t) = \frac{\partial M(\mathbf{r}, t)}{\partial t} \quad (1)$$

subject to a partially absorbing boundary condition at the interface between the pore fluid and the solid grain

$$[D_0 \hat{\mathbf{n}} \cdot \nabla M(\mathbf{r}, t) + \rho M(\mathbf{r}, t)]_{\text{interface}} = 0. \quad (2)$$

Here D_0 is the diffusion coefficient for the bulk fluid and the parameter ρ describes interface absorption. (In Eq. (1) the effect of bulk relaxation has already been factored out [9].) This Brownstein-Tarr formalism ignores the effects of field inhomogeneities due to magnetic susceptibility contrasts. In practical terms this means the theory may be limited to low field NMR [9].

The value of ρ is generally larger for transverse relaxation than for longitudinal relaxation but, in both cases, is assumed to be constant over the entire interface. Equations (1) and (2)

*Corresponding author: johnson10@slb.com

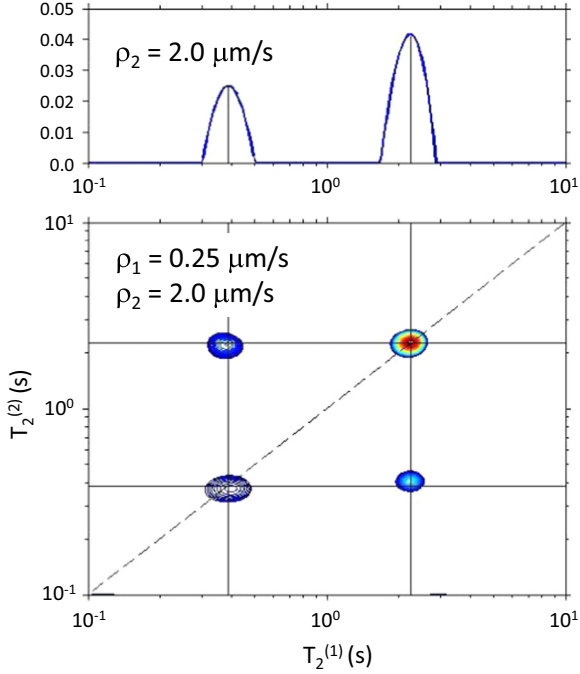


FIG. 1. (Color online) One- and two-dimensional lifetime distributions obtained from our numerical simulations. The storage time, $t_s = 3$ s, and location of the peaks in the 2D distribution are seen to be directly related to locations of their 1D counterparts. The assumed surface relaxivity values are indicated.

can be solved directly for very simple geometries or by random walk simulations for more complex pore models. In a recent paper, we applied variable step size random walk techniques to the μ – GC model to study the T_2 -storage- T_2 process in a system with two quite different pore sizes [8]. Our purpose was to see if our results could be understood in terms of the empirical two-site exchange model, which is often employed to interpret 2D NMR data. The calculations presented there were limited to the case in which there was no relaxation during the storage period but were sufficient to expose a number of significant limitations of the two-site exchange model.

Because the controlling equations (1) and (2) are derived from a self-adjoint operator, we know that the eigenmodes of the system form a complete and orthonormal set [10]. Several authors have extended the formalism originally developed by Brownstein and Tarr [9] to the analysis of 2D NMR [8,11,12]. From the outset one must consider two sets of eigenfunctions and eigenvalues. We denote these as $\{\mathbf{q}_n(\mathbf{r})\}$ and $\{\lambda_n\}$ for transverse decay and $\{\mathbf{Q}_n(\mathbf{r})\}$ and $\{\mu_n\}$ for longitudinal decay. The orthonormality condition is simply

$$\begin{aligned} \mathbf{q}_i \cdot \mathbf{q}_j &= \delta_{ij} \quad i, j = 1, 2, \dots, \\ \mathbf{Q}_k \cdot \mathbf{Q}_l &= \delta_{kl} \quad k, l = 1, 2, \dots, \end{aligned} \quad (3)$$

where the inner product of two functions is $\mathbf{f} \cdot \mathbf{g} = \int_{V_P} \mathbf{f}(\mathbf{r}) \mathbf{g}(\mathbf{r}) d\mathbf{r}$.

In the 1D case the total magnetic moment can be represented as a series expansion. For T_2 decay,

$$M_2(t) = M_0 V_P \sum_{n=1}^{\infty} A_n e^{-\lambda_n t}, \quad (4)$$

where M_0 is the initial magnetization and V_P is the volume of the pore space. We have $A_n = (\mathbf{m}_0 \cdot \mathbf{q}_n)^2 / V_P$, where $\mathbf{m}_0(\mathbf{r})$ is a function equal to unity in the pore space and zero elsewhere. Similarly, the T_1 decay function is

$$M_1(t) = M_0 V_P \sum_{n=1}^{\infty} B_n e^{-\mu_n t}, \quad (5)$$

where $B_n = (\mathbf{m}_0 \cdot \mathbf{Q}_n)^2 / V_P$. It is understood that T_1 is conventionally monitored with an inversion-recovery procedure in which the quantity actually measured is $M_0 V_P - 2M_1(t)$, but we will refer to $M_1(t)$ as the T_1 decay function.

The theoretical description of the T_2 -storage- T_2 experiment begins, as in Eq. (4), with an expansion of $M(\mathbf{r}, t)$ in terms of $\{q_n, \lambda_n\}$ during the encoding period $0 \leq t \leq \tau_1$. $M(\mathbf{r}, \tau_1)$ is then the initial condition for an expansion, covering the storage period, based on the longitudinal eigenfunctions and eigenvalues, $\{\mathbf{Q}_l\}$ and $\{\mu_l\}$. [Because $M(\mathbf{r}, \tau_1)$ will in general not be spatially uniform, its projection onto the longitudinal eigenfunctions is non-trivial.] Finally, $M(\mathbf{r}, \tau_1 + t_s)$ is the initial state for the transverse basis function expansion between $t = \tau_1 + t_s$ and $t = \tau_1 + t_s + \tau_2$, where τ_2 is the length of the detection period. The result is [11,12]

$$M(\tau_1, \tau_2 : t_s) = \sum_{m,n} \mathcal{M}_{nm}(t_s) e^{-[\lambda_n \tau_1 + \mu_m \tau_2]}, \quad (6)$$

where

$$\mathcal{M}_{nm}(t_s) = M_0 \sum_l (\mathbf{m}_0 \cdot \mathbf{q}_n)(\mathbf{q}_n \cdot \mathbf{Q}_l)(\mathbf{Q}_l \cdot \mathbf{q}_m)(\mathbf{q}_m \cdot \mathbf{m}_0) e^{-\mu_l t_s}. \quad (7)$$

From Eq. (7) we see, by inspection, that the matrix is symmetric: $\mathcal{M}_{nm}(t_s) = \mathcal{M}_{mn}(t_s)$. Furthermore, each coefficient of $e^{-\mu_l t_s}$ is positive for the diagonal elements, M_{ii} . Finally, we note a useful sum rule which follows from the completeness of the set $\{\mathbf{q}_n\}$:

$$\sum_{n,m} \mathcal{M}_{nm}(t_s) = M_0 \sum_l (\mathbf{m}_0 \cdot \mathbf{Q}_l)^2 e^{-\mu_l t_s} \equiv M_1(t_s). \quad (8)$$

Thus, if one sums over all elements of the storage matrix (integrates over all the decay times in the contour plot) the result must lie on the T_1 decay curve. This result provides a powerful check on experimental data and, as we shall see, on the results of our own numerical simulations.

It is worth pointing out that if ρ_1 and ρ_2 are small enough, no matter how complicated the geometry as long as the pore space is connected, then there is only one decay rate each for T_1 and for T_2 and there are no off-diagonal elements in the 2D experiment. (There is only the one diagonal peak.) Moreover, if ρ_1 and ρ_2 are large enough such that the slow diffusion limit applies for T_1 and for T_2 there will be multiple decay rates for each but there will be no off-diagonal elements in a T_2 -storage- T_2 experiment simply because the normal modes for T_1 and for T_2 are identical. So this off-diagonal effect

can happen only if at least one of T_1 and T_2 are intermediate between fast and slow diffusion, however that is defined.

The above equations are quite general. A tremendous simplification of this formalism occurs if we can make two related assumptions, viz (1) that two modes are sufficient to describe both the T_1 and T_2 decay spectra and (2) that any relevant function of position within the pore space can be expressed as a linear combination of these two orthonormal basis functions (e.g., $\mathbf{q}_1$ and $\mathbf{q}_2$ or, equivalently, $\mathbf{Q}_1$ and $\mathbf{Q}_2$). [In general, continuum models of porous media (like the GC or μ - GC models) have a countably infinite number of normal modes [9]; in most cases only a few contribute to any NMR signal. However, it is possible to quantify these higher order modes by means of direct numerical computations on model porous media [13].] These assumptions allow us to derive an analytic expression for the matrix elements $\mathcal{M}_{nm}(t_s)$ as we now demonstrate.

Because $(\mathbf{f} \cdot \mathbf{g})^2 \leq (\mathbf{f} \cdot \mathbf{f})(\mathbf{g} \cdot \mathbf{g})$ for any $\mathbf{f}, \mathbf{g}$ pair and because $|\mathbf{m}_0| = \sqrt{\mathbf{m}_0 \cdot \mathbf{m}_0} = \sqrt{V_P}$, we may define an angle ψ via

$$\mathbf{m}_0 \cdot \mathbf{q}_1 = \sqrt{V_P} \cos(\psi) = \sqrt{V_P A_1}. \quad (9)$$

The orthonormality and completeness of $\mathbf{q}_1$ and $\mathbf{q}_2$ implies $\mathbf{m}_0 = (\mathbf{m}_0 \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{m}_0 \cdot \mathbf{q}_2)\mathbf{q}_2$ which, in turn, implies $(\mathbf{m}_0 \cdot \mathbf{q}_1)^2 + (\mathbf{m}_0 \cdot \mathbf{q}_2)^2 = \mathbf{m}_0 \cdot \mathbf{m}_0 = V_P$. We then have $|\mathbf{m}_0 \cdot \mathbf{q}_2| = \sqrt{V_P} \sin(\psi) = \sqrt{V_P A_2}$. Similarly, we may define the angle θ by writing $\mathbf{Q}_1$ in terms of $\mathbf{q}_1$ and $\mathbf{q}_2$:

$$\mathbf{Q}_1 = \cos(\theta)\mathbf{q}_1 + \sin(\theta)\mathbf{q}_2. \quad (10)$$

Since $\mathbf{Q}_1 \cdot \mathbf{Q}_2 = 0$, we have

$$\mathbf{Q}_2 = \pm[\sin(\theta)\mathbf{q}_1 - \cos(\theta)\mathbf{q}_2]. \quad (11)$$

Substituting these results into Eq. (7), the matrix elements $\mathcal{M}_{ij}$ can be written as

$$\begin{aligned} \mathcal{M}_{11}/(M_0 V_P) &= \cos^2(\psi)[\cos^2(\theta)e^{-\mu_1 t_s} + \sin^2(\theta)e^{-\mu_2 t_s}], \\ \mathcal{M}_{22}/(M_0 V_P) &= \sin^2(\psi)[\sin^2(\theta)e^{-\mu_1 t_s} + \cos^2(\theta)e^{-\mu_2 t_s}], \\ \mathcal{M}_{12}/(M_0 V_P) &= (1/4)\sin(2\psi)\sin(2\theta)[e^{-\mu_1 t_s} - e^{-\mu_2 t_s}]. \end{aligned} \quad (12)$$

The six integrals implied by Eq. (7) have been reduced to two parameters, ψ and θ .

Because the longitudinal decay can also be expressed in terms of just two distinct normal modes, we can define an angle χ in terms of the projection of the initial uniform magnetization onto the T_1 basis functions:

$$\cos(\chi) \equiv \mathbf{m}_0 \cdot \mathbf{Q}_1 / \sqrt{V_P}. \quad (13)$$

In view of Eq. (10), the angles ψ , θ and χ are related by

$$\cos(\chi) = \cos(\theta)\cos(\psi) + \sin(\theta)\sin(\psi) = \cos(\theta - \psi). \quad (14)$$

In a similar fashion we may write, utilizing Eq. (11),

$$\sin(\chi) = |\mathbf{m}_0 \cdot \mathbf{Q}_2| / \sqrt{V_P} = |\sin(\theta - \psi)|. \quad (15)$$

Either of Eq. (14) or (15) implies

$$\theta = \psi \pm \chi. \quad (16)$$

The foregoing analysis lends itself to a simple visual depiction, which we show in Fig. 2. Consider any one of the four

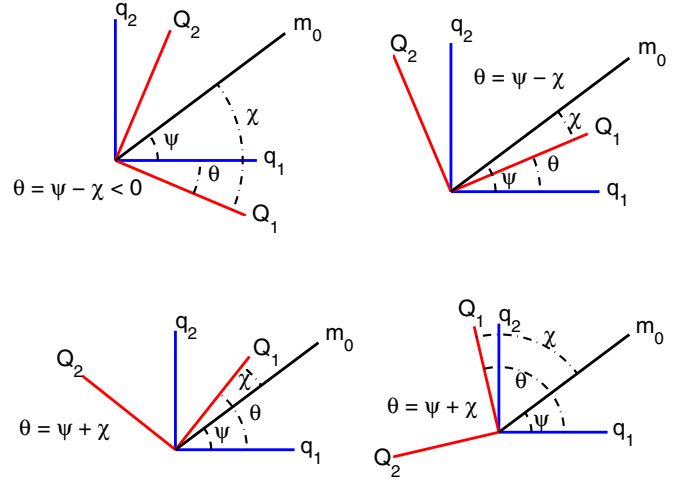


FIG. 2. (Color online) Schematic plot of the possible relationships between the two orthogonal modes for T_2 decay, $\mathbf{q}_{1,2}$, the two orthogonal modes for T_1 decay, $\mathbf{Q}_{1,2}$, and the initial uniform magnetization, $\mathbf{m}_0$, within the context of the present theory. We imagine that the orientation of $\mathbf{Q}_{1,2}$ to $\mathbf{q}_{1,2}$ is continuously rotated counterclockwise in going from the upper-left picture to the lower-right.

subplots. The T_2 normal modes, $\mathbf{q}_1$ and $\mathbf{q}_2$, are orthonormal. Any point on the graph represents a function which is a linear combination of these two functions, the coefficients of which are the coordinates on the graph. In particular, for porous media being considered in this article the initial uniform magnetization, $\mathbf{m}_0$, can be represented in terms of its coordinates along these two axes, thus defining the angle ψ as shown. Similarly, each of the two functions $\mathbf{Q}_1$ and $\mathbf{Q}_2$, which are mutually orthonormal, may also be represented as points in the $\mathbf{q}_1, \mathbf{q}_2$ plane. This is the basic assumption of the article. Their coordinates define the angle θ , as shown. The four subplots indicate how the picture evolves from left to right, top to bottom, as θ is continuously increased. It also shows each possibility of Eq. (16). Depending upon which side of $\mathbf{m}_0$ the vector $\mathbf{Q}_1$ lies, either of the $\pm$ signs in Eq. (16) may hold, in principle. In most cases of interest $\rho_1 \leq \rho_2$ and we believe that the minus sign in this equation is appropriate. When $\rho_1 = 0$ the mode $\mathbf{Q}_1$ corresponds to uniform magnetization, so $\chi = 0$ and $\theta = \psi$. When $\rho_1 = \rho_2$ one has $\mathbf{Q}_i \equiv \mathbf{q}_i$, so $\theta = 0$. For the most common situation encountered experimentally we have $0 < \rho_1 < \rho_2$, and therefore $0 < \theta < \psi$ which implies $\theta = \psi - \chi$.

Our algorithm, then, consists of the following steps: Fit the T_2 decay to the sum of two simple exponentials, with amplitudes $A_1 = \cos^2(\psi)$ and $A_2 = 1 - A_1$, and decay rates λ_1 and λ_2 . Solve for the angle $\psi = \cos^{-1}(\sqrt{A_1})$. Likewise, fit the T_1 decay to the sum of two simple exponentials, with amplitudes $B_1 = \cos^2(\chi)$ and $B_2 = 1 - B_1$, and decay rates μ_1 and μ_2 . Solve for the angle $\chi = \cos^{-1}(\sqrt{B_1})$; take $\theta = \psi - \chi$ and compute the storage matrix elements using Eqs. (12).

In the very special case that the two subsystems are unconnected, as happens for example if there is a barrier between them [8], then the amplitudes for both M_1 and

M_2 decay are simply the respective macro- and micropore volumes, i.e., $A_i = B_i$ $i = 1, 2$. In this very special case, then, $\psi \equiv \chi$ and $\theta \equiv 0$. The result according to Eqs. (12) is that $M_{12} \equiv 0$ and the two diagonal elements are simply the respective single exponential 1D M_1 decays, as one would intuitively expect.

Even if the 1D NMR relaxation experiments can be represented accurately as a sum of two exponentials it is possible for our theory of 2D experiments to fail. Consider any subplot in Fig. 2. Let us imagine that there is a third mode each for T_1 and T_2 relaxation, $\mathbf{Q}_3$ and $\mathbf{q}_3$, respectively. In Fig. 2 each points in a direction perpendicular to the plane of the figure. In this situation neither of these modes contributes to their respective 1D decays nor does either contribute to the 2D results. If, however, we imagine that the triplet $\{\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3\}$ is rotated rigidly around the vector $\mathbf{m}_0$ by an amount β , keeping the angle χ fixed and maintaining their mutual orthonormality we see that the T_1 decay is still represented as a sum of two exponentials because this third mode does not couple to the initial magnetization; the third mode is still orthogonal to the initial magnetization, $\mathbf{m}_0 \cdot \mathbf{Q}_3 \equiv 0$, for any value of β . This, then, would be a situation in which each 1D relaxation can be represented as a sum of two exponential decays but the storage matrices, $\mathcal{M}_{ij}$, require three exponentials because $\mathbf{Q}_3 \cdot \mathbf{q}_j \neq 0$. The sum rule, Eq. (8), still holds, however, and it has only two terms, the coefficients of the third exponential having canceled out.

These ideas can also be applied to the T_1 - T_2 protocol in which spins undergo longitudinal decay up to a switching time, T_s , and are then tipped into the transverse plane for an additional time τ . One has

$$M(T_s; \tau) = \sum_{n,m} \mathcal{M}_{nm}^{T_1, T_2} e^{-\mu_n T_s} e^{-\lambda_m \tau}, \quad (17)$$

where $\mathcal{M}_{nm}^{T_1, T_2} = M_0(\mathbf{m}_0 \cdot \mathbf{Q}_n)(\mathbf{Q}_n \cdot \mathbf{q}_m)(\mathbf{q}_m \cdot \mathbf{m}_0)$. Although the results could be presented in terms of the angles ψ and χ , they can more simply and succinctly be written in terms of the T_1 and T_2 amplitudes directly as

$$\frac{\mathcal{M}_{nm}^{T_1, T_2}}{M_0 V_p} = B_n A_m + (2\delta_{nm} - 1)\sqrt{A_1 A_2 B_1 B_2}. \quad (18)$$

In Ref. [12] very general arguments were used to conclude that negative off-diagonal matrix elements are to be expected in the analysis of T_1 - T_2 data. Equation (18) provides a specific example of this. Of the two off-diagonal elements, $\mathcal{M}_{12}^{T_1, T_2}$ and $\mathcal{M}_{21}^{T_1, T_2}$, one is negative and the other is positive. (A trivial exception occurs when $A_1 = B_1$, in which case both off-diagonal elements are zero.) The accuracy of the above equations is being pursued and the results will be presented elsewhere [15].

B. Numerical calculations

We confirm the validity of the theory by means of a comparison against a series of numerical simulations based on the μ -GC model [8] a schematic of which is shown in Fig. 3. The model consists of a periodic arrangement of overlapping grains each of which is comprised of smaller grains. Our calculations are based on variable step size random walks as described in detail in [8]. We have employed the

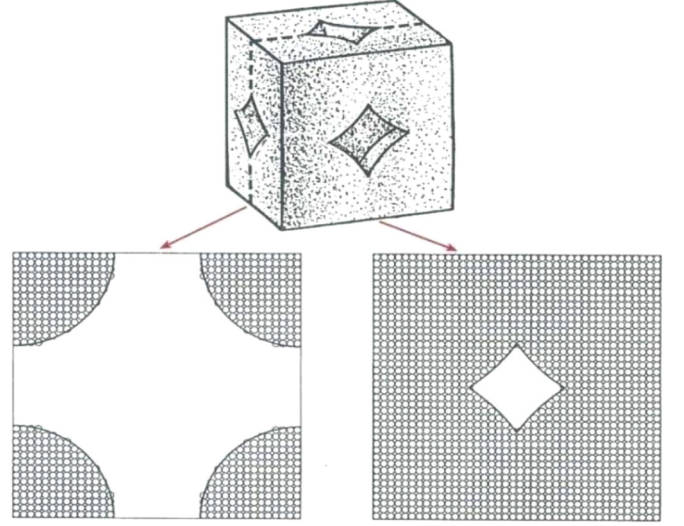


FIG. 3. (Color online) A microporous version of the grain consolidation model in which the large grains are comprised of smaller grains whose centers lie on a lattice constructed by subdividing the original unit cell. The solid curves separate the macroporous and the microporous regions.

same geometry as in [8] but with varying values of the surface relaxivities, ρ_1 and ρ_2 . The probability that a walker will be removed from the count when it hits the pore-grain interface is simply related to the values of these physical parameters.

For most of the calculations presented here the pore space is divided into two distinct, but coupled, populations whose dimensions differ by roughly a factor of 20 (see Table I of Ref. [8]). We see, however, that the two peaks in the T_2 distribution shown in the upper panel of Fig. 1 are separated by less than a factor of 10. Before going further, let us explain why this is the case. It is straightforward to estimate the position of the short-time peak in the distribution since the micrograins are simply smooth overlapping spheres. The pore volume to surface area ratio for this part of the pore space is $0.839 \mu\text{m}$ and the corresponding lifetime can be estimated as $T_2 = V_p/(S\rho_2) = 0.42$ s. This is what we see in Fig. 1. If the macropores had smooth surfaces, their V_p/S ratio would be $15.2 \mu\text{m}$ and the corresponding lifetime would be 7.6 s, which is well above the value seen in Fig. 1. Of course, the macropores do *not* have smooth surfaces; the interface between the two pore populations is controlled by the large amount of surface area associated with the micrograins. This interface is where the diffusive coupling takes place; its presence is thus quite apparent in just the T_2 distribution.

TABLE I. Amplitudes and decay times deduced by fitting two exponentials to numerical simulations of $M(t)$. It is understood that when a specific value of ρ is used for the storage part of the simulation, the entries for λ_i now represent μ_i and that for A_1 represents B_1 .

$\rho(\mu\text{m/s})$	0	0.25	0.5	1.0	2.0
$1/\lambda_1(\text{s})$	∞	8.2	5.05	3.3	2.25
A_1	1.0	0.924	0.825	0.735	0.620
$1/\lambda_2(\text{s})$	3.5	1.7	1.2	0.61	0.42

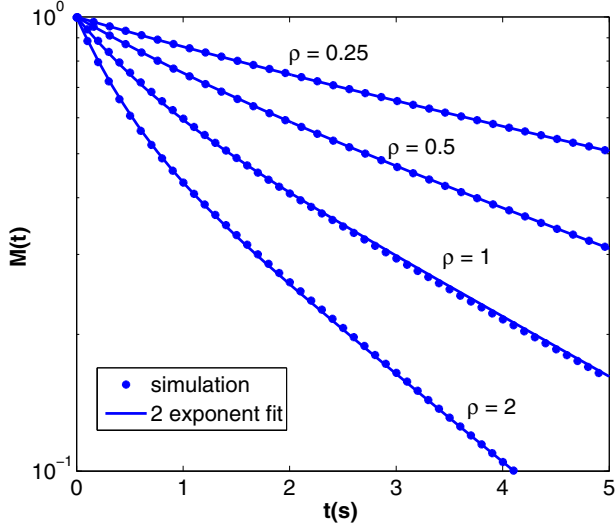


FIG. 4. (Color online) Comparison of the results of numerical simulations of the 1D decays, $M_i(t)$ $i = 1, 2$, for different values of the surface relaxivity, ρ , in $\mu\text{m/s}$.

A necessary condition for the success of this theory is that each of the 1D measurements, $M_1(t)$ and $M_2(t)$, must be accurately represented as a sum of two exponential functions. To test this part we have performed a number of simulations, assuming different values of relaxivity, ρ , whose results are shown in Fig. 4. We see that each decay is well represented as a sum of two exponentials. The fitting parameters are collated in Table I and are plotted in Fig. 5. The decay rate μ_2 for the case where $\rho = 0$ was determined by means of a separate simulation, in which walkers were initially released only in the macropore and the number of walkers remaining in the macropore as a function of time was monitored [8].

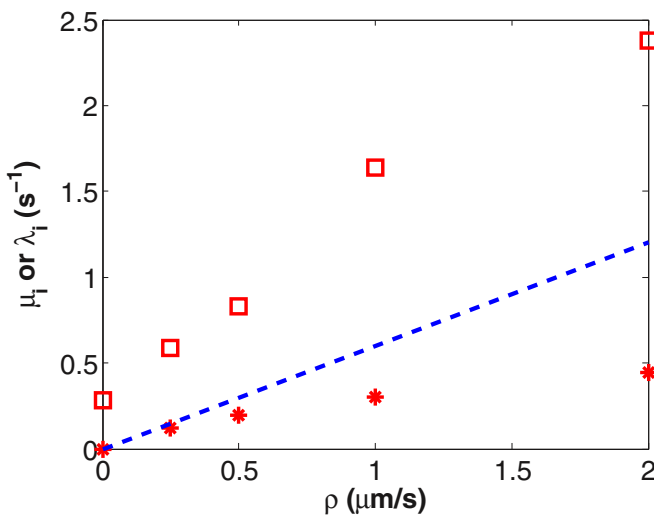


FIG. 5. (Color online) The two decay rates deduced from our 1D simulations as a function of surface relaxivity. For the 2D simulations the decay rates for $\rho = 2 \mu\text{m/s}$ are appropriate to the T_2 decay parts; the others are appropriate to T_1 decay, i.e., to the storage part of the simulations. The dashed line is Eq. (19) using the known surface-to-pore-volume ratio.

On quite general grounds [9] the slowest relaxation rate in the limit of small ρ is

$$\lim_{\rho \rightarrow 0} \mu_1(\rho) = \rho(S/V_P), \quad (19)$$

where (S/V_P) is the total surface area to pore volume ratio of the pore system. From the geometry of our model we have determined that $(S/V_P) = 0.603 (\mu\text{m})^{-1}$. Equation (19) is plotted as a dashed line in Fig. 5 using this value. In this limit, the slowest mode, $\mathbf{q}_1$ or $\mathbf{Q}_1$ as the case may be, corresponds to uniform magnetization throughout the entire pore system. The normalization condition, Eq. (3), requires that this value is equal to $\frac{1}{\sqrt{V_P}}$ throughout the entire pore system.

We note that the second decay rate does *not* approach zero as $\rho \rightarrow 0$ but it tends to a constant, which constant is directly proportional to the bare diffusion constant. That is because this mode corresponds to a distribution which has one sign in the middle of the macropore and the opposite sign in the micropores. This fact follows immediately from the orthogonality condition, Eq. (3), on $\mathbf{Q}_2$ which reads

$$0 = \mathbf{Q}_1 \cdot \mathbf{Q}_2 \equiv \frac{1}{\sqrt{V_P}} \int_{V_P} Q_2(\mathbf{r}) dV. \quad (20)$$

Relaxation for this mode occurs strictly by the diffusion between the macro- and the micropores so as to even out the imbalance. Equation (20) is the reason why a separate simulation [8] was needed in order for us to deduce the decay rate of this $\mathbf{Q}_2$ mode; it does not couple to an initial condition of uniform magnetization. In general, then, except by accident the ratio of the decay rates of the two modes is essentially never equal to the ratio of the pore sizes (20 in this article) no matter the value of the surface relaxivity, ρ . Diffusive coupling is an integral part of each of the modes. The slowest mode does *not* correspond to magnetization exclusively in

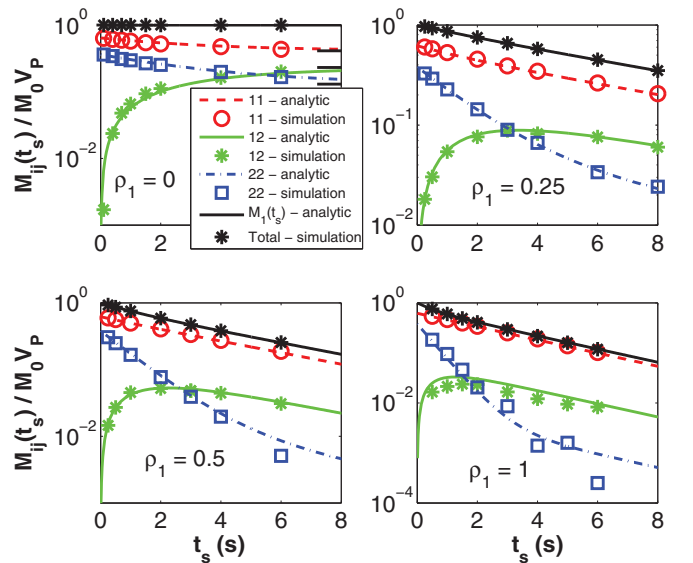


FIG. 6. (Color) The results of numerical simulations of the 2D storage matrix, $M_{ij}(t_s)$, in the micro grain consolidation model are compared against the predictions of the analytic model, Eqs. (12). The values assumed for ρ_1 , in $\mu\text{m/s}$, are indicated. In all cases we used $\rho_2 = 2 \mu\text{m/s}$.

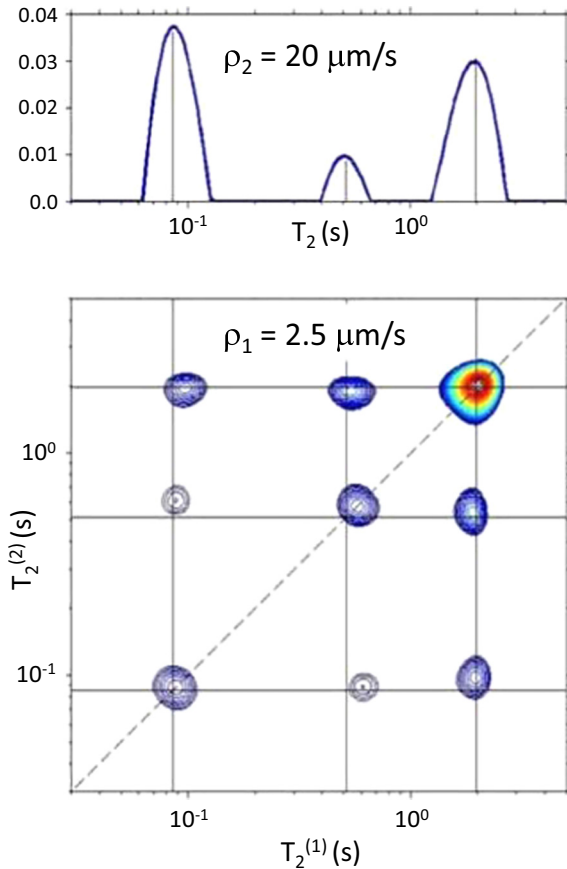


FIG. 7. (Color online) Numerical results for the 2D storage matrix, $M_{ij}(t_s)$, in a micrograin consolidation model in which the sizes of the micro- and macrograins have been increased by a factor of 2 relative to our previous calculations. The storage time is $t_s = 2$ s the values of ρ_1 and ρ_2 are given in the figure.

the macropores; the second mode does *not* correspond to magnetization exclusively in the micropores. Rather, each mode is a distribution throughout the entire pore space.

Numerical simulations of T_2 -storage- T_2 were carried out using the parameters $\rho_2 = 2 \mu\text{m/s}$ and $\rho_1 = 0, 0.25, 0.5$, and $1.0 \mu\text{m/s}$. A large number of random walkers (typically 10^6) are launched randomly and uniformly within the pore space. For the initial part of the simulation, $t \leq \tau_1$, relaxation at the pore-grain interface is controlled by ρ_2 . At the first switching time, τ_1 , the surviving walkers continue but with a surface relaxation rate, ρ_1 . Then at the second switching time, $\tau_1 + t_s$, the surface relaxivity reverts to ρ_2 . We processed the time

domain data using an inverse 2D Fredholm integral equation solver to generate two dimensional T_2 - T_2 contour plots, one of which is shown in Fig. 1. We integrated the volume under each peak in order to determine M_{ij} for each value of the storage time, t_s . The results are shown in Fig. 6 along with the analytic results of Eqs. (12) using as input data the parameters listed in Table I. We see that the theory works quite well for this geometry and for these choices of relaxivity values [14]. We also see that the simulation data obeys the sum rule Eq. (8).

In principle, this comparison could have been made based on just the time domain data using the definition, Eq. (6) and the model, Eqs. (12). This would have involved the analysis of a large number of 1D data sets (one for each value of τ_1 and t_s). We employed the 2D Fredholm equation solver as a matter of convenience; its use is not essential.

Obviously, there are many conditions under which it will not be possible to express both of the 1D decays as the sum of two exponentials. In such cases we want to understand what features of the 2D NMR spectrum contain information that cannot be derived from the 1D distributions. An example can be seen in the μ - GC model if we enlarge the unit cell size by a factor of two and increase the values of ρ_1 and ρ_2 as shown in Fig. 7. In this case the 1D decays can be written as the sum of three exponentials and it is possible to extend the present theory [15]. In more complicated systems, our analysis will inevitably breakdown and we will be in a position to identify those features of the 2D spectra that truly contain new information about diffusive coupling.

III. CONCLUSIONS

In summary, if the 1D decay functions, $M_1(t)$ and $M_2(t)$, can be represented accurately as the sum of two exponentials, we have developed an analytic theory for the results of T_1 - T_2 and T_2 -storage- T_2 experiments. The only inputs to the theory are the decay rates and amplitudes of the two 1D measurements. The validity of the theory has been confirmed by comparison against numerical modeling in a mixed pore size system. While the present analysis is based on the diffusion equation and fluid filled porous media, our results apply equally to measurements on surfactants and catalysts where the evolution of the magnetization is governed by a different self-adjoint operator. The only other requirement is that the initial magnetization and the final amplitude measurement are described by the same function, $\mathbf{m}_0$.

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- [14] For $\rho_1 = 1 \mu\text{m/s}$, the values of $\mathcal{M}_{12}$ calculated from the simulations do not agree with the analytic theory as well as in the other three cases. The larger the value of ρ_1 , the smaller are the off-diagonal peaks; they must vanish as $\rho_1 \rightarrow \rho_2$. Our guess is that this lack of agreement reflects limitations in the ability of the inverse Fredholm solver to accurately describe the very small off diagonal peaks rather than limitations in the accuracy of Eqs. (12).
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